

NILADRI PAL

AI/ML Engineer • Full-Stack Data Solutions • Automation & Cloud Systems

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PROFESSIONAL SUMMARY

Versatile AI & Software Systems Engineer with an MCA degree, offering multi-disciplinary expertise across Data Analytics, Data Science, Machine Learning Engineering, AI Automation, and Full-Stack Web Development. Skilled in architecting automated ETL data ingestion pipelines (cutting overhead by 35%), training verified ML models (>85% accuracy), orchestrating LLM/RAG workflows with high-dimensional vector embeddings, developing containerized REST APIs (FastAPI), and deploying production web systems across PaaS and cloud platforms. Proven aptitude in building scalable, production-ready software solutions.

TECHNICAL SKILLS & COMPETENCIES

Languages & Scripting: Python (Advanced), SQL (PostgreSQL, MySQL, SQLite), JavaScript (ES6+), Bash/Shell Scripting, HTML5, CSS3

AI, GenAI & Automation: LLM Orchestration (LangChain, LlamaIndex), RAG Architecture, Vector DBs (FAISS, ChromaDB), Prompt Engineering, PyTorch, Scikit-Learn, NLP, Computer Vision, Feature Engineering, Workflow Automation

Web & Backend Engineering: FastAPI, Flask, RESTful API Design, Asynchronous Execution (asyncio), Microservices Architecture, JSON Streaming, Postman API Testing

Data Analytics & Databases: Pandas, NumPy, Advanced SQL (Window Functions, CTEs, Joins, Indexing, Query Tuning), Automated ETL Pipelines, Outlier Imputation, Model Evaluation (MSE, RMSE, R²), Exploratory Data Analysis (EDA), Time Series Modeling

Cloud, DevOps & Deployment: Docker (Containerization), PaaS & Production Hosting (Render, Vercel, Netlify), Google Cloud Run, Git/GitHub (CI/CD Workflows)

Systems & Productivity: Linux (Kali/Ubuntu System Admin), Bash Shell Scripting, Microsoft Excel (Advanced Pivot Tables, Lookups, Data Modeling, AI Excel Tools & Copilot Automation)

PROFESSIONAL EXPERIENCE

AI/ML & Data Trainee — Euphoria GenX

Jan 2026 – Jun 2026 | Kolkata, India

- Automated Data Pipelines:** Architected automated Python ETL pipelines for multi-source data ingestion, eliminating manual intervention and slashing preprocessing overhead by 35%.
- ML Baselines & Validation:** Developed and tuned supervised predictive models using Scikit-Learn; maintained accuracy consistently above 85% via k-fold cross-validation and hyperparameter optimization.
- Exploratory Data Analysis & APIs:** Conducted statistical validation and anomaly detection on high-dimensional datasets; packaged trained inference routines into modular REST API endpoints.

Cybersecurity Intern — Ardent Computech Pvt. Ltd.

Mar 2025 – Jul 2025 | Kolkata, India

- Log Telemetry Analytics:** Analyzed structured Linux system log streams (syslog, auth.log) and high-throughput network telemetry to detect operational anomalies, performance bottlenecks, and security events.
- Compliance Automation:** Authored custom Python automation utilities to extract, structure, and sanitize endpoint telemetry for automated audit reporting; hardened Linux configurations.

TECHNICAL PROJECTS

Smart Study Assistance & Text Retrieval System [ai-study-assistant-1-lwsb.onrender.com]

Python, LangChain, FastAPI, NLP, Render

- Engineered an intelligent study assistant utilizing RAG architecture, semantic chunking, and vector indexing for context-aware academic content retrieval.
- Built an asynchronous FastAPI backend delivering sub-second response times (<200ms); containerized and deployed on Render Cloud via GitHub CI/CD.

Urban-Twin: Multi-Camera Traffic Analytics [urbantwin-ai.onrender.com]

Python, Spatial Tracking, Computer Vision, Cloud Dashboard

- Designed real-time streaming analytics pipelines calculating vehicle velocities, congestion density, and wait times; hosted a live tracking dashboard on Render.

Earth & Satellite Imagery Analytics Platform [earth-and-satellite-1.onrender.com]

Python, Computer Vision, Geospatial Data, FastAPI, Render

- Processed multi-spectral satellite imagery and spatial rasters to detect land-cover transitions, vegetation indices (NDVI), and topographical variations; deployed on Render to serve live inference via REST APIs.

Retail Sales Forecasting & Demand Analytics

Python, Time Series, Scikit-Learn, Pandas, Feature Engineering

- Modeled multi-year retail purchasing trends using rolling window aggregations and lag features; minimized RMSE to optimize supply inventory replenishment.

Predictive Real Estate Valuation System

Python, Scikit-Learn, Pandas, Statistical Modeling, REST API

- Executed full EDA, outlier treatment, and categorical encoding across 10,000+ records; trained regularized regression models achieving an R² score of 0.88.

EDUCATION

Master of Computer Applications (MCA) — Future Institute of Engineering and Management

2024 – 2026 | Kolkata, WB

Bachelor of Computer Applications (BCA) — Raiganj University

2019 – 2022 | West Bengal, India

CERTIFICATIONS & ACHIEVEMENTS

- NPTEL Certification: Data Analytics Using Python** — Validated expertise in Pandas, NumPy, and statistical machine learning.
- Smart India Hackathon (SIH) Participant:** Developed real-time data analytical solutions within an intensive 36-hour sprint.
- Leadership & Coordination: Cultural Club Organizing Committee** — Managed event logistics, operational budgets, and technical resources.